

Dipak Sumesara

Senior Backend Engineer

Node.js | TypeScript | Express.js | MongoDB | Microservices | AWS

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SUMMARY

Senior Backend Engineer with 8+ years of experience and a core stack of Node.js, TypeScript, and MongoDB, building scalable microservices and event-driven systems on AWS. For 3 years, architected and led the backend of a US enterprise SaaS platform, cutting API response times from seconds to milliseconds. Built real-time backend features for a fantasy sports platform serving 100K+ concurrent users at peak, and backend services for a Singapore government health platform with a 90% test-coverage gate. Also builds LLM applications (RAG, agents, guardrails) in Node.js and TypeScript.

TECHNICAL SKILLS

- **Backend:** Node.js, TypeScript, JavaScript, Express.js, NestJS, Fastify, GraphQL, RESTful APIs
- **Databases:** MongoDB, PostgreSQL, MySQL, Redis (caching, sessions), Elasticsearch, Schema Design, Data Modeling
- **Architecture:** System Design, Distributed Systems, Microservices, Service Decomposition, Event-Driven Architecture, API Gateway, Pub/Sub, Message Queues (SQS, DLQ, BullMQ), Asynchronous Processing, Caching Strategies, Real-Time Systems (WebSockets), High Concurrency, Fault-Tolerant Design, API Versioning, API Contracts
- **Reliability and Operations:** Scalability, High Availability, Failover, Load Balancing, Monitoring, Logging, Alerting (Sentry), Incident Management and Response, Root Cause Analysis (RCA), Rollbacks, Hotfix Deployment
- **Cloud and DevOps:** AWS (Lambda, API Gateway, SQS, S3, CloudFront), Serverless, Docker, CI/CD
- **Security:** Authentication and Authorization, Session Management, Secure API Design, Centralized Auth
- **Testing:** Jest, Vitest (unit and integration), High Test-Coverage Gates, Error Handling, Performance Profiling
- **AI Engineering:** Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), OpenAI API, Anthropic Claude API, Open-Source LLMs (Llama, Qwen via Ollama), Embeddings, Vector Search (cosine similarity), Agentic Workflows, Multi-Step Agent Orchestration, Tool Use and Function Calling, Conversation Memory, Guardrails, Prompt Engineering, Context Window Management, Structured Output Validation (Zod), Streaming Responses (SSE)
- **AI Dev Tools:** GitHub Copilot, Cursor, Claude Code, OpenAI Codex
- **Frontend:** React, Redux Toolkit, Angular, Ionic

PROFESSIONAL EXPERIENCE

SwagUp

Mar 2023 - Apr 2026

Senior Backend Engineer / Platform Lead | Remote

Tech Stack: Node.js, Express.js, TypeScript, MongoDB

Aloha: Enterprise Admin and Operations Platform

- Architect and lead for a B2B enterprise operations platform (vendor, distributor, and admin workflows); owned system design, built the backend, and led the team, working alongside US product and engineering teams.
- Decomposed the platform into microservices behind an API Gateway, with centralized authentication, authorization, and session management on AWS Lambda.
- Designed schemas and data models, versioned APIs, and enforced API contracts through specifications, schema enforcement, and response transformers.
- Cut API response times from seconds to milliseconds via Redis caching, query optimization, and database indexing.
- Scaled for high availability with connection pooling, read/write splitting, replication, failover, and load balancing.
- Implemented a reliable Node.js-to-Salesforce data sync on AWS SQS, with a dead-letter queue (DLQ) to capture and recover failed messages.
- Designed fault-tolerant, event-driven flows and asynchronous job processing on pub/sub, SQS, and BullMQ, with retry and observability patterns.
- Developed Elasticsearch-backed search for ~20K products with custom indexing pipelines that kept the index consistent with source data.
- Handled production incidents with Sentry, logging, and alerting, including root cause analysis, rollbacks, hotfix deployments, and automatic recovery.
- Containerized services with Docker for Kubernetes-based deployments and owned CI/CD pipelines, branching strategy, and release workflows, including off-hours production deployments under high-availability SLAs.
- Established Jest test automation and code-review standards across the platform.

SwagUp.com: Customer-Facing Platform

- Built REST and GraphQL APIs for order tracking and a multi-tier pricing engine with real-time quote generation and dynamic discount logic.

CoffeeBeans Consulting

Jun 2021 - Mar 2023

Software Engineer / Backend Lead | Remote

Tech Stack: Node.js, Express.js, TypeScript, MongoDB, PostgreSQL

Health Promotion Board (Healthy 365): Singapore Government Health Platform

- Built Node.js backend systems for Healthy 365, a citizen health app, in a 10-person cross-functional team.
- Delivered under enforced quality gates (90% minimum unit-test coverage, commits blocked below threshold, strict ESLint rules) with pair programming, dual QA, and staged Dev, QA, UAT, and Production promotion.

Salam Kisan: Agritech Platform

- Owned backend architecture on Strapi CMS for an agricultural supply-chain platform; built REST APIs, Open edX integration for learning content, and automated data-sync workflows.
- Led a 3-engineer backend team across sprint delivery, code quality, and production support.

Yudiz Solutions

May 2020 - May 2021

Software Engineer | Ahmedabad, India

Tech Stack: Node.js, Express.js, TypeScript, MongoDB, React

11Wickets: Fantasy Sports Platform (Dream11 competitor)

- Built real-time backend features for a platform serving 100K+ concurrent users at peak match load, using WebSockets for live score updates and Redis caching to sustain high-concurrency read traffic.
- Implemented WebSocket-based live chat for real-time engagement during matches, and mentored junior developers.

Biztech Consulting & Solutions

Jan 2019 - Mar 2020

Software Engineer | Ahmedabad, India

Tech Stack: TypeScript, MySQL

- Proposed and led a v2.0 overhaul, migrating the product from legacy Angular to a modern, modular architecture with clean design patterns and secure API integrations; adopted by the founding team.
- Rebuilt the product from the ground up as independent modules, enabling tiered pricing (Professional, Ultimate, and Custom) in place of the all-or-nothing model and driving significant customer growth.

Angular Minds

Dec 2017 - Sep 2018

Software Engineer (Intern to Full-Time) | Pune, India

Tech Stack: Node.js, NestJS, Express.js, TypeScript, MongoDB, Angular, Ionic

- Built Numnu, a food-event discovery app with a NestJS and MongoDB backend and an Angular/Ionic frontend, surfacing events by proximity and date range, plus an event-guide mode used at the Royal Manitoba Winter Fair.

PROJECTS

AI Commerce Assistant

github.com/dipak-sumesara/ai-commerce-assistant

Tech Stack: Node.js, TypeScript, Fastify, Ollama (Qwen, nomic-embed-text), Zod, React, Vitest

- Built a RAG pipeline that chunks a markdown knowledge base, generates embeddings, and retrieves the top-K chunks by cosine similarity (semantic search) from an in-memory vector store to ground answers.
- Converted natural-language shopping queries into Zod-validated filters executed deterministically in TypeScript, behind a scope-classifier guardrail that blocks off-topic requests before retrieval or generation.
- Streamed responses via SSE; tested guardrails, prompt injection, structured output, and vector search with Vitest.

AI Integration Discovery Agent

github.com/dipak-sumesara/ai-integration-discovery-agent

Tech Stack: Node.js, TypeScript, Express.js, Anthropic Claude API, Ollama, Zod, Vitest

- Built an agentic LLM workflow that reads a third-party API's documentation, uses tool calling to probe the running API (authentication, users, groups, roles, scopes), and emits a Zod-validated connector specification.
- Wrote the multi-step agent loop without a framework, with swappable LLM providers (Anthropic Claude, Ollama).

Reverie: AI Character Chat

github.com/dipak-sumesara/ai-chat-app

Tech Stack: Node.js, TypeScript, Express.js, MongoDB, Redis, Ollama (Llama), React

- Built conversation memory that keeps long chats within the model's context window and cuts prompt token usage: a recent-message window plus AI-generated summaries cached in Redis and refreshed in the background.
- Streamed replies token by token over Server-Sent Events (SSE); stored editable, branching chat history in MongoDB.

EDUCATION

Master of Computer Applications (MCA) | Savitribai Phule Pune University, Pune

2018

Bachelor of Computer Applications (BCA) | Krantiguru Shyamji Krishna Verma Kachchh University, Bhuj

2015